

Khai Loong Aw

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Education

2024 - now	Stanford University , <i>Ph.D. Computer Science</i> Building world models that learn vision like a human baby.
2020 - 2024	Singapore Management University , <i>B.Sc. Computer Science</i> GPA: 4.0/4.0.

Selected Publications (* = equal contribution)

2025	Taming generative world models for zero-shot optical flow extraction S Kim*, KL Aw* , K Kotar*, C Eyzaguirre, W Lee, Y Liu, J Watrous, S Stojanov, JC Niebles, J Wu, DLK Yamins NeurIPS 2025
2024	Instruction-tuning Aligns LLMs to the Human Brain KL Aw , S Montariol, B AlKhamissi, M Schrimpf, A Bosselut COLM 2024
2023	Training language models to summarize narratives improves brain alignment KL Aw , M Toneva ICLR 2023 - Spotlight
2022	Detecting False Alarms from Automatic Static Analysis Tools H.J. Kang, K.L. Aw , D. Lo ICSE 2022 - Distinguished Paper Nominee

Research Experience

2024 - now	Stanford , <i>CS PhD student</i> • Working with Prof. Daniel Yamins and Prof. Michael Frank • Train multi-billion parameter AI models from scratch on distributed GPUs, cloud TPUs • Building visual-cognitive world models for AI, cognitive science, and neuroscience
2023	EPFL , <i>Research Intern (NeuroAI)</i> • Advisors: Martin Schrimpf, Antoine Bosselut • Investigated the similarities between: (1) LLM representations and human brain fMRI activity, and (2) LLM and human behavior. Showed that LLM-brain alignment strongly correlates with model size ($r = 0.95$) and world knowledge ($r = 0.81$). [COLM 2024] • Contributed LLaMA and T5 models to the open-science Brain-Score Language platform
2022	Max Planck Institute for Software Systems , <i>Research Intern (NeuroAI)</i> • Advisor: Mariya Toneva • Designed a novel interpretability method to evaluate brain alignment across various narrative elements (e.g., characters, emotions). Showed that training language models to summarize narratives improves alignment to human brain fMRI activity [ICLR spotlight]
2022	Singapore Management University , <i>Research Assistant (Computer Vision)</i> • Advisor: Sun Qianru • Used Self-Supervised Learning as a pre-training step to improve the performance of existing methods for the computer vision task of Semantic Segmentation
2021	Software Analytics Research Group , <i>Research Intern (Machine Learning)</i> • Advisor: David Lo • Used AI interpretability methods to identify data leakage issues with state-of-the-art ML models for detecting software bugs. [ICSE Distinguished Paper Nominee]

- 2017 **National University of Singapore, Research Assistant**
- Advisor: Dan Zhao
 - Synthesized novel metal-organic framework materials for carbon capture

Industry Experience

- 2023 **GIC Singapore, NLP Project**
- Built pipeline of text-to-SQL and retrieval-augmented generation LLMs, allowing employees to use English questions to query global stock market databases
- 2022 **GovTech Singapore, AI Engineer Intern (Computer Vision)**
- Improved the processing speed of state-of-the-art algorithms for Multi-Object Tracking, i.e., tracking the motion of multiple objects or humans within a video sequence

Presentations

- 2025 **Talk**, Princeton HassonLab's 1kD meeting
- 2025 **Talk**, Stanford Language and Cognition Lab
- 2025 **Talk**, Stanford NeuroAI Lab
- 2023 **Oral**, ICLR 2023
- 2023 **Talk**, ML Tea @ Max Planck Institute for Software Systems

Awards

- 2024 - 2025 **Stanford School of Engineering Fellowship**
- 2024 **IMDA Gold Medal Award** - top student in B.Sc. CS graduating cohort
- 2022 **Best Poster Presentation**, MPI-SWS Institute Retreat 2022
- 2022 **Outstanding Second Year Student Award** - top student in Year 2 of CS undergrad
- 2021 **Outstanding First Year Student Award** - top student in Year 1 of CS undergrad
- 2020 **Global Impact Scholarship** - the university's top full-ride scholarship

Hackathons

- 2021 **1st place**, SMU Legal Innovation and Technology (LIT) Hackathon
- 2021 **3rd place**, SMU Ellipsis Tech Series Hackathon presented by Goldman Sachs
- 2021 **Dentons Rodyk Access to Justice Award**, SMU LIT Hackathon
- 2021 **People's Choice Award**, Asia-Pacific Legal Innovation and Technology Association
- 2021 **Tech-cess to Justice Award**, Asia-Pacific Legal Innovation and Technology Association
- 2021 **Legal Tech Entrant Award**, Asia-Pacific Legal Innovation and Technology Association
- 2020 **Best College Hack**, Hack-a-Solution international hackathon

Service

- 2023 - now **Reviewer**
- NeurIPS 2024 (Top Reviewer), 2025
- Queer in AI Workshop @ NeurIPS 2024, NAACL 2024
- Workshop on Representational Alignment @ ICLR 2024
- Workshop on Unifying Representations in Neural Models @ NeurIPS 2023
- Workshop on Instruction Tuning and Instruction Following @ NeurIPS 2023
- 2021 - now **Mentor**
- Stanford CS PhD Student-Applicant Support Program (SASP) 2024
- Stanford CS Undergraduate Mentoring 2024
- Stanford Graduate Application Mentorship Program (GradAMP) 2024
- QueerInAI Graduate Application Mentorship Program 2024
- SMU "Pay It Forward" Mentorship Program 2024
- SMU Data Associate Program 2021, 2023 - mentored eight students on their ML projects
- 2021 - 2022 **Vice-President, SMU BIA**
- Led team of 23 directors to organize Machine Learning workshops & events for a club with 1000+ students; allow non-computing students to gain hands-on ML experience

2020 - 2022 **Organizer, SMU BIA**
Workshop on ML Deployment
Workshop on Deep Learning in NLP
Workshop on Interview Preparation for ML and Data Engineers
Workshop on Interview Preparation for Business and Data Analysts
Other events: Data Science CTF Hackathon, Industry Mentor Sharing Session, Project Polish Program, and projects between SMU BIA and industry partners